



Started on	Thursday, 17 September 2026, 9:19 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:31 AM
Time taken	11 mins 25 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  #include<stdlib.h>
3
4  int findFloor(int arr[], int low, int high, int x) {
5      int floor_val = -1;
6
7      while (low <= high) {
8          int mid = low + (high - low) / 2;
9
10         if (arr[mid] == x) {
11             return arr[mid];
12         }
13
14         if (arr[mid] < x) {
15             floor_val = arr[mid];
16             low = mid + 1;
17         }
18         else {
19             high = mid - 1;
20         }
21     }
22
23     return floor_val;
24 }
25
26 int main() {
27     int n;
28
29     if (scanf("%d", &n) != 1) {
30         return 0;
31     }
32
33     int *arr = (int*)malloc(n * sizeof(int));
34     if (arr == NULL) {
35         return 1;
36     }
37
38     for(int i = 0; i < n; i++) {
39         scanf("%d", &arr[i]);
40     }
41
42     int x;
43
44     if (scanf("%d", &x) != 1) {
45         free(arr);
46         return 0;
47     }
48
49     int result = findFloor(arr, 0, n - 1, x);
50     printf("%d\n", result);
51
52     free(arr);

```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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